Schedules

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Session Objective

- Schedules
- Schedules based on recoverability
- Schedules based on Serializability

Session Outcome

At the end of this session, participants will be able to

- Understand the scheduling
- Understand the schedule based on recoverability
- Understand the schedule based on Serializability

Schedules

Schedule or History: A schedule (or history) S of n transactions $T_1, T_2, ..., T_n$ is an **ordering of the operations** of the transactions subject to the constraint that:

- For each transaction T_i that participates in S, the operation of T_i in S must appear in the same order in which they occur in T_i
- Operations from different transactions can be interleaved in the schedule (such that the operations from transactions T_j can be interleaved with the operations of T_i in S).

The order of operations in S is considered to be **total ordering**.

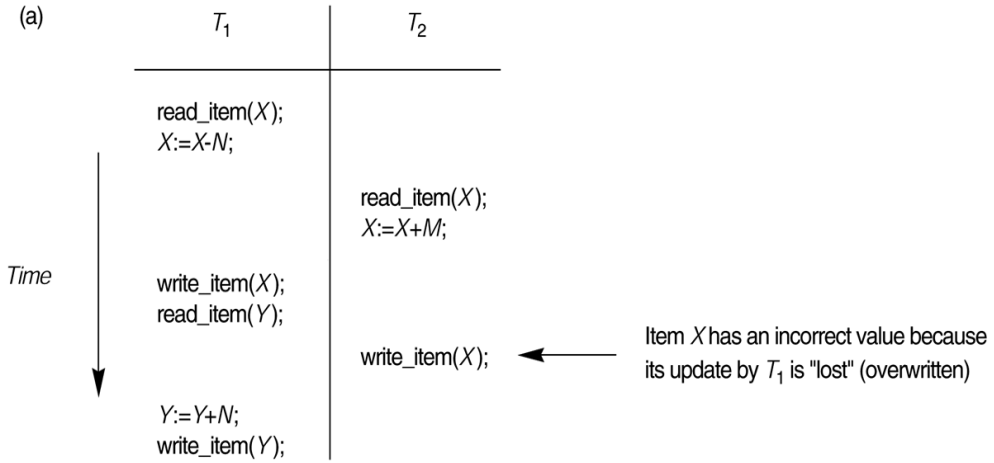
- For any two operations in the schedule, one must occur before the other.

Schedules

- Schedules can also be displayed in more compact notation
- Order of operations from left to right
- Include only read (r) and write (w) operations, with transaction id (1, 2, ...) and database item name (X, Y, ...)
- Can also include other operations such as b (begin), e (end), c (commit), a (abort)

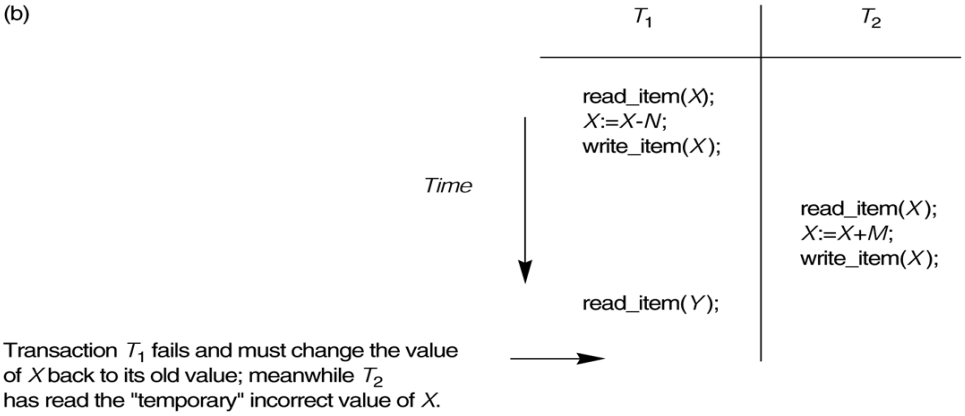
Eg: $S_a : r1(X); r2(X); w1(X); r1(Y); w2(X); w1(Y);$

Schedules - Example1



$$S_a : r_1(X); r_2(X); w_1(X); r_1(Y); w_2(X); w_1(Y);$$

Schedules - Example2



Assume the transaction T_1 is aborted

$S_b : r_1(X); w_1(X); r_2(X); w_2(X); r_1(Y); a_1$

Conflicting Operations in a Schedule

Two operations in a schedule are said to be **conflict** if they satisfy all the three conditions:

- If they belong to different transactions
- If they access the same item X
- At least one of the operations is a write_item(X)

Conflicting and Non_Conflicting Operations in a Schedule

$S_a : r_1(X); r_2(X); w_1(X); r_1(Y); w_2(X); w_1(Y);$

Conflicting Operations:

- $r_2(X)$ and $w_1(X)$
- $r_1(X)$ and $w_2(X)$
- $w_1(X)$ and $w_2(X)$

Non_Conflicting Operations:

- $r_1(X)$ and $r_2(X)$
- $w_2(X)$ and $w_1(Y)$
- $r_1(X)$ and $w_1(Y)$

Conflicting Operations in a Schedule

Two operations conflict if changing their order results in a different outcome

- **Read_Write Conflict** : Changing $r_1(X); w_2(X)$ to $w_2(X); r_1(X)$ means that T1 will read a different value for X
- **Write_Write Conflict**: Changing $w_1(Y); w_2(Y)$ to $w_2(Y); w_1(Y)$ means that the final value for Y in the database can be different
- **Read operations are not conflicting**: Changing $r_1(Z); r_2(Z)$ to $r_2(Z); r_1(Z)$ does not change the outcome

Complete Schedule

A schedule S of n transactions $T_1, T_2, T_3.....T_n$ is said to be **complete schedule** if the following conditions hold:

- The operation in S are exactly those operations in $T_1, T_2, T_3.....T_n$ including commit or abort operation as the last operation
- For any pair of operations from the same transaction T_i, their relative order of appearance in S is the same as their order of appearance in T_i
- For any two conflicting operations, one of the two must occur before the other in the schedule

Characterizing Schedules Based on Recoverability

Schedules are characterized as follows:

- **Recoverable schedule:**
  - Once a transaction T is committed, it should never be necessary to roll back (T)
  - This ensures that the durability property of transactions is not violated.
  - A schedule S is **recoverable** if no transaction T in S commits until all transactions T' that have written some item X that T reads have committed.

T1	T2
$W_1(X)$	
	$W_2(X)$
	Commit $T_2(X)$
	(Waits for T1 to commit first)
Commit $T_1(X)$	

Table 1: Recoverable

Characterizing Schedules Based on Recoverability

- **Non-Recoverable schedule:**
  - A schedule where a committed transaction may have to be rolled back during recovery called as non-recoverable schedule.
  - This violates durability from ACID properties (a committed transaction cannot be rolled back) and so non-recoverable schedules should not be allowed.

T1	T2
$W_1(X)$	
	$W_2(X)$
	Commit $T_2(X)$
Abort $T_1(X)$	

Table 2: UnRecoverable

T2 commited cannot able to Rollback so unrecoverable schedules

Characterizing Schedules Based on Recoverability

Consider two schedules:
$Sc : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); c_2; a_1;$
Sc is non-recoverable

$Sd : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); w_1(Y); c_1; c_2;$
Sd is recoverable
Sc is not recoverable, because T2 reads item X from T1, and then T2 commits before T1 commits. If T1 aborts after the C2;

Then the value of X that T2 read is no longer valid and T2 must be aborted after it had already committed $- - - >$ **schedule not recoverable.**

For the schedule to be **recoverable** the c_2 operation is postponed until after T_1 commits
$Sc1 : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); c_1; c_2;$
In recoverable schedule no committed transactions ever needs to be rolled back

Cascading Rollback

An uncommitted transaction has to be rolled back because it read an item from a transaction that failed. This phenomenon is known as **Cascading rollback**

$Se : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); w_1(Y); a_1; a_2;$

The transaction T_2 has to be rolled back because it read item X from T_1 which is failed transactions and T_1 then aborted

Cascading roll back is time consuming process since numerous transactions to be rolled back.

Cascadless

A schedule is said to be a **Cascadless schedule**, or to avoid cascading rollback, if every transaction in the schedule **reads only items** that were written by committed transactions.

$Se : r_1(X); w_1(X); r_2(X); r_1(Y); w_2(X); w_1(Y); a_1; a_2;$

To avoid rollback :
Then $r_2(X)$ must be postponed untill after T_1 has committed (aborted), thus delaying T_2.

Strict Schedule

- A schedule in which a transaction $T2$ can either **read or write** an item X until the transaction $T1$ that last wrote X has committed
- Simplify the recovery process by simply undoing a write_item(X) of an aborted transaction to restore the before image of data item X.
- **Sf:W1(X,5);W2(X,8);a1**
- if T1 aborts, the recovery procedure recovers the before image which was orginally 9, even it has been changed to 8 by transaction T2.
- S_f is not **strict scheule** since it permits T2 to write item X even though the last wrote X had not commited

Schedule - Example

Schedule C: $r_1(X); w_1(X); r_1(Y); w_1(Y); c_1; r_2(X); w_2(X);$
Schedule C is **cascadeless and also strict**

Schedule D: $r_1(X); w_1(X); w_3(X); r_1(Y); w_1(Y); c_1; r_2(X); w_2(X);$
Schedule D is cascadeless, but not strict (which writes the value of X before T1 commits)

Schedule E: $r_1(X); w_1(X); r_1(Y); w_1(Y); c_1; w_3(X); r_2(X); w_2(X);$
Schedule E is neither cascadeless nor strict, to make it strict and cascadless, w2(X) and r2(x) must be delayed until after T3 commits

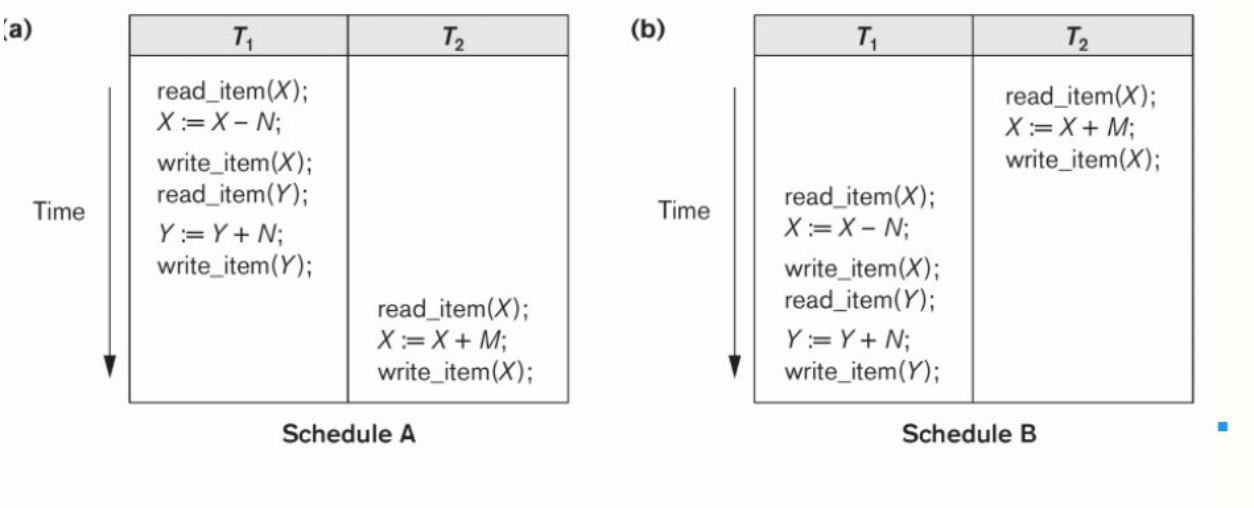
Summary

- The set of all possible schedules can be partitioned into two subsets: **recoverable and non-recoverable**
- A **cascadeless** schedule will be the subset of the **recoverable** schedules.
- A **strict schedules** will be the subset of **cascadless** schedules.
- All strict schedules are cascadless and all cascadless schedules are recoverable.

Serializability of Schedules

- Consider two transactions T1 and T2 which is submitted at the same time.
  If no interleaving is permitted then there are two possible ways:
  - Execute all the operations of T1 and then T2
  - Execute all the operations of T2 and then T1
- **Serializability theory**, attempts to determine which schedules are "correct" and which are not and to develop techniques that allow only correct schedules

Serial Schedules



Schedule A and B are called as Serial Schedule

Serial Schedules

- Schedules A and B are called serial schedules because the operations of each transaction are executed consecutively, without any interleaved operations from the other transaction.
- In a serial schedule:
  - Transactions are performed in serial order
  - Only one transaction is active at any time
  - Commit or abort of the active transactions initiates execution of next transactions
  - Since transactions are independent **every serial schedule produce correct result**

Serial Schedules: Disadvantage

- Serial schedules are not feasible for performance reasons:No interleaving of operations.
- Long transactions force other transactions to wait.
- System cannot switch to other transaction when a transaction is waiting for disk I/O or any other event.
- Need to allow concurrency with interleaving without sacrificing correctness

Non - Serial Schedules

Time

T_1	T_2
read_item(X);	
X:=X-N;	
	read_item(X);
	X:=X+M;
write_item(X);	
read_item(Y);	
	write_item(X);
Y:=Y+N;	
write_item(Y);	

Schedule C

T_1	T_2
read_item(X);	
X:=X-N;	
write_item(X);	
	read_item(X);
	X:=X+M;
	write_item(X);
read_item(Y);	
Y:=Y+N;	
write_item(Y);	

Schedule D

Two nonserial schedules C and D with interleaving of operations

Serial & Non - Serial Schedules

X=90 and Y=90, N=3 and M=2 serial Schedule A, B expect database values X=89 and Y=93. C produces the value X=92 and Y=93

Time	T_1	T_2
	read_item(X);	
	X:=X-N;	
		read_item(X);
		X:=X+M;
	write_item(X);	
	read_item(Y);	
		write_item(X);
	Y:=Y+N;	
	write_item(Y);	

Schedule C

T_1	T_2
read_item(X);	
X:=X-N;	
write_item(X);	
	read_item(X);
	X:=X+M;
	write_item(X);
read_item(Y);	
Y:=Y+N;	
write_item(Y);	

Schedule D

Serial & Non - Serial Schedules

X=90 and Y=90, N=3 and M=2 serial Schedule A, B expect database values X=89 and Y=93. D produces the value X=89 and Y=93

(a)

Time	T_1	T_2
	read_item(X);	
	$X := X - N$;	
	write_item(X);	
	read_item(Y);	
	$Y := Y + N$;	
	write_item(Y);	
		read_item(X);
		$X := X + M$;
		write_item(X);

Schedule A

(b)

Time	T_1	T_2
		read_item(X);
		$X := X + M$;
		write_item(X);
	read_item(X);	
	$X := X - N$;	
	write_item(X);	
	read_item(Y);	
	$Y := Y + N$;	
	write_item(Y);	

Schedule B

Characterizing Schedule based on Serializability

Schedules C and D are nonserial because of interleaving operations from two transactions.

Some non-serial schedules give correct results.

We have to determine which of the non-serial schedule always give a correct result.

A schedule S of n transactions is **serializable** if it is equivalent to some serial schedule of the same n transactions.

Question is : When are two schedules considered "equivalent" ?

- Result equivalent
- Conflict equivalent - Conflict Serializable
- View equivalent - View Serializable

Result Equivalent

- Two schedules are called result equivalent if they produce the same final state of the database.
- Disadvantages:
  - Not all the result equivalent schedules produces same final state.
  - Cannot be used to define equivalence of schedules.

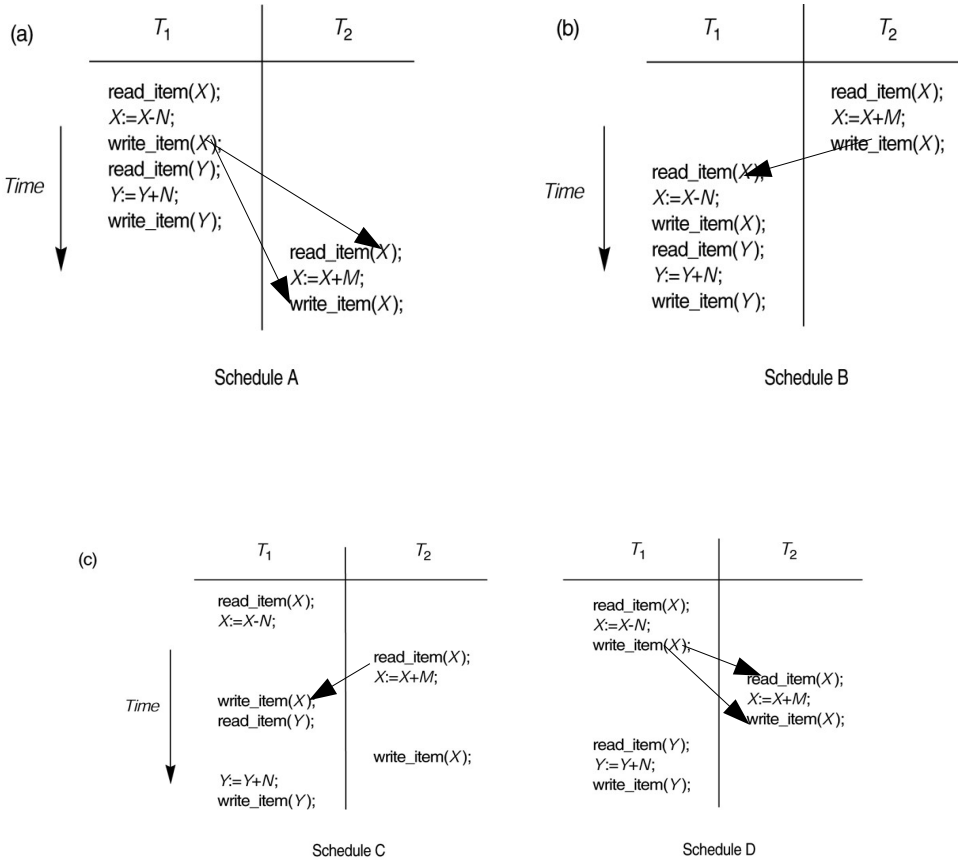
S_1	S_2
read_item(X);	read_item(X);
X:=X+10;	X:=X*1.1;
write_item(X);	write_item(X);

Two schedules that are result equivalent for the initial value of X = 100 but are not result equivalent in general.

Conflict Equivalent

- Two schedules are said to be **conflict equivalent** if the order of any two conflicting operations is the same in both schedules.
- According to this, schedule D is equivalent to the serial schedule A. Since A is serial schedule and D is equivalent to A, **D is serializable schedule.**
- We say that a schedule S is conflict serializable if it is conflict equivalent to a serial schedule, hence D is conflict serializable.
- Conflict equivalence can be obtained using
  - Ordering of conflict operations
  - Swapping of non-conflict operations
  - Precedence Graph

Conflict Equivalent - Ordering of Conflict operations



Conflict Equivalent - Swapping of non - conflict operations

T_1	T_2
read(A)	
$A := A - 50$	
write (A)	
read(B)	
$B := B + 50$	
write(B)	
	read(A)
	$temp := A * 0.1$
	$A := A - temp$
	write(A)
	read(B)
	$B := B + temp$
	write(B)

S_1

S_1

T_1	T_2
read(A)	
write(A)	
	read(A)
	write(A)
read(B)	
write(B)	
	read(B)
	write(B)

S_2

Conflict Equivalent - Swapping of non-conflict operations

If a schedule S can be transformed into an equivalent schedule S' by a series of swaps of non-conflicting instructions, then S and S' are conflict serializable.

T_1	T_2
read(A)	
write(A)	
	read(A)
	write(A)
read(B)	
write(B)	
	read(B)
	write(B)

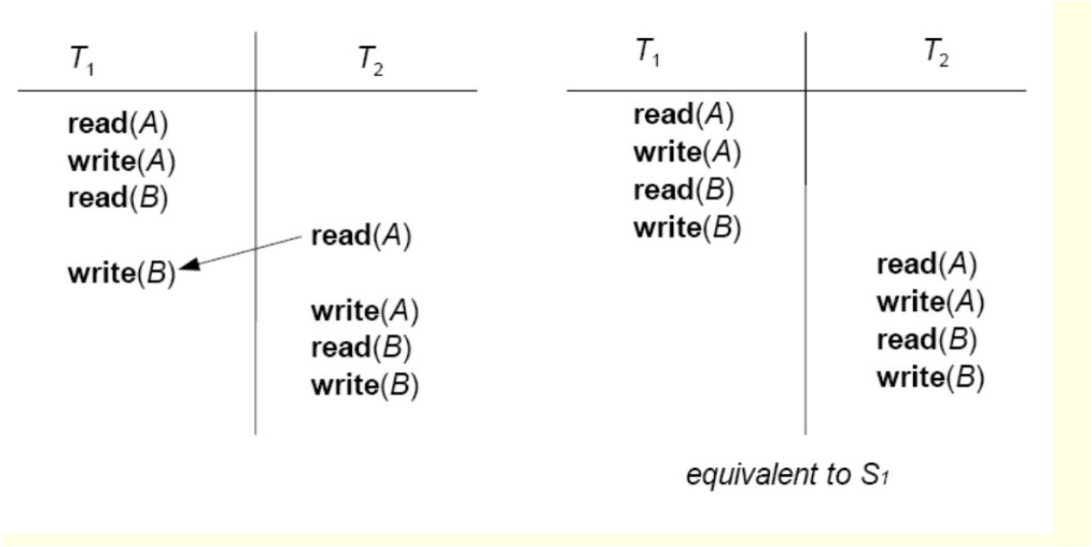
T_1	T_2
read(A)	
write(A)	
	read(A)
read(B)	
	write(A)
write(B)	
	read(B)
	write(B)

Conflict Equivalent

T_1	T_2
read(A)	
write(A)	
	read(A)
read(B)	
	write(A)
write(B)	
	read(B)
	write(B)

T_1	T_2
read(A)	
write(A)	
read(B)	
	read(A)
	write(A)
write(B)	
	read(B)
	write(B)

Conflict Equivalent



Conflict Equivalent - Precedence Graph

- The algorithm looks at only the read_item and write_item operations in a schedule to construct a precedence graph (or serialization graph),
- The graph is a directed graph $G = (N, E)$ that consists of a set of nodes $N = T_1, T_2, ..., T_n$ and a set of directed edges $E = e_1, e_2, ..., e_m$.
- There is one node in the graph for each transaction T_i in the schedule.
- Each edge e_i in the graph is of the form $(T_j \rightarrow T_k)$, where T_j is the starting node of e_i and T_k is the ending node of e_i.
- Edge from node T_j to node T_k is created by the algorithm if a pair of conflicting operations exist in T_j and T_k
- The conflicting operation in T_j appears in the schedule before the conflicting operation in T_k .

Testing for conflict serializability

- For each transaction T_i participating in s Schedule S, create a node labeled T_i in the precedence graph.
- For each case an edge is created from T_i to T_j if one of the operations in T_i appears before a conflicting operation in T_j
  - When T_j executes a read_item(X) after T_i executes a write_item(X), create an edge (T_ißT_j) in the precedence graph.
  - where T_j executes a write_item(X) after T_i executes a read_item(X), create an edge (Ti $\rightarrow$ Tj) in the precedence graph.
  - when T_j executes a write_item(X) after T_i executes a write_item(X), create an edge (T_ißT_j) in the precedence graph.
- The schedule is **serializable** if and only if the precedence graph has no cycles.

Serial & Non - Serial Schedules

T_1	T_2
read_item(X);	
X:=X-N;	
	read_item(X);
	X:=X+M;
write_item(X);	
read_item(Y);	
	write_item(X);
Y:=Y+N;	
write_item(Y);	

Time

Schedule C

T_1	T_2
read_item(X);	
X:=X-N;	
write_item(X);	
	read_item(X);
	X:=X+M;
	write_item(X);
read_item(Y);	
Y:=Y+N;	
write_item(Y);	

Schedule D

Two nonserial schedules C and D with interleaving of operations

(a)

T_1	T_2
read_item(X);	
$X := X - N$;	
write_item(X);	
read_item(Y);	
$Y := Y + N$;	
write_item(Y);	
	read_item(X);
	$X := X + M$;
	write_item(X);

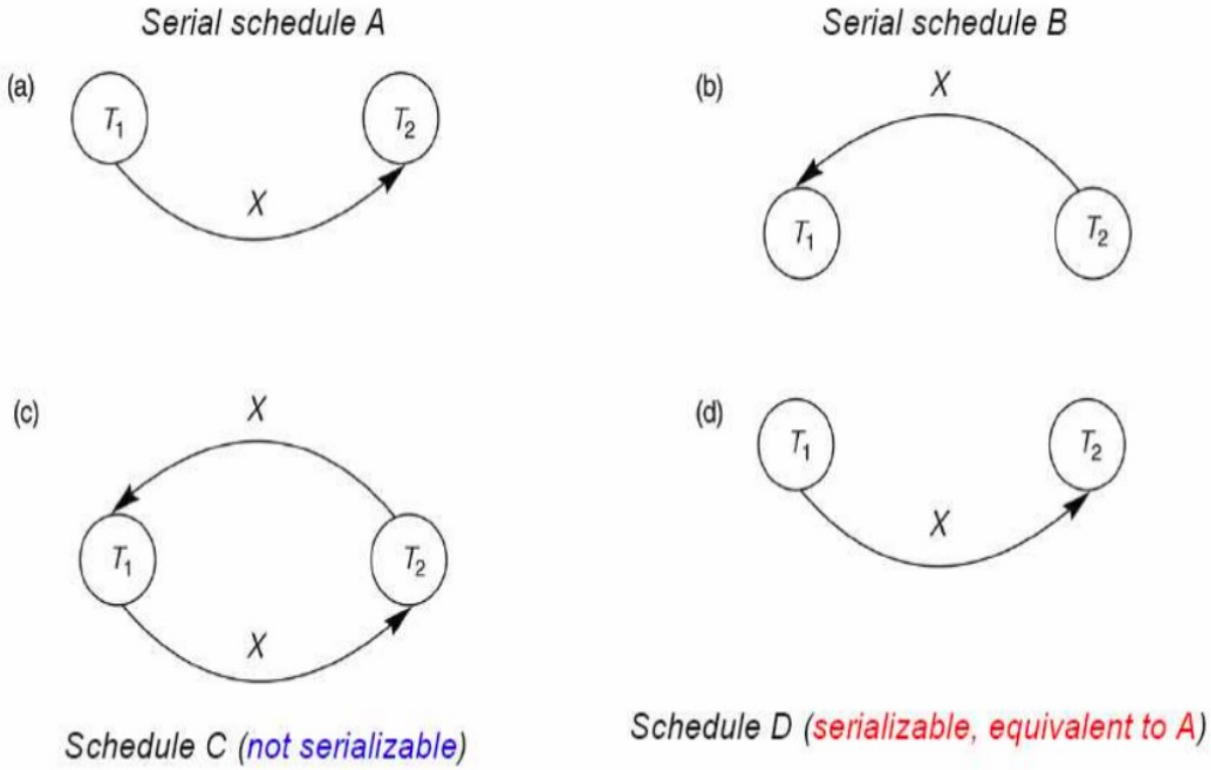
Time

(b)

T_1	T_2
	read_item(X);
	$X := X + M$;
	write_item(X);
read_item(X);	
$X := X - N$;	
write_item(X);	
read_item(Y);	
$Y := Y + N$;	
write_item(Y);	

Time

Precedence Graph



Precedence Graph - Example

(b)

Time ↓

Transaction T_1	Transaction T_2	Transaction T_3
	read_item(Z);	
	read_item(Y);	
	write_item(Y);	
		read_item(Y);
		read_item(Z);
read_item(X);		
write_item(X);		
		write_item(Y);
		write_item(Z);
	read_item(X);	
read_item(Y);		
write_item(Y);	write_item(X);	

Schedule E

(c)

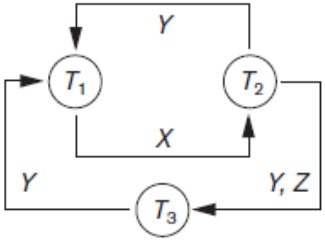
Time ↓

Transaction T_1	Transaction T_2	Transaction T_3
		read_item(Y);
		read_item(Z);
read_item(X);		
write_item(X);		
		write_item(Y);
		write_item(Z);
	read_item(Z);	
read_item(Y);		
write_item(Y);		
	read_item(Y);	
	write_item(Y);	
	read_item(X);	
	write_item(X);	

Schedule F

Precedence Graph - Example

(d)



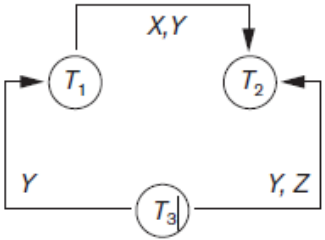
Equivalent serial schedules

None

Reason

Cycle $X(T_1 \rightarrow T_2), Y(T_2 \rightarrow T_1)$
Cycle $X(T_1 \rightarrow T_2), YZ\ (T_2 \rightarrow T_3), Y(T_3 \rightarrow T_1)$

(e)



Equivalent serial schedules

$T_3 \rightarrow T_1 \rightarrow T_2$

View Equivalent

- A less restrictive definition of equivalence of schedules is called view equivalence.
- A schedule is view serializable if it is view equivalent to a serial schedule.
- Schedules are said to be view equivalent if the following three conditions hold:
  - The same set of transactions participates in S and S' and S and S' includes same operations of those transactions.
  - For any operation, $R_i(X)$ of T_i in S, if the value of X read was written by an operation W_j(X) of T_j (or if it is the original value of X before the schedule started), the same condition must hold for the value of X read by operation R_i(X) of Ti in S'. (Initial read and update read)
  - If the operation $W_k(Y)$ of T_k is the last operation to write item Y in S, then $W_k(Y)$ of T_k must also be the last operation to write item Y in S'. **(Final write)**

View Serializable

- As long as each read operation of a transaction reads the result of the same write operation in both schedules.
- The write operations of each transaction must produce the same results.
- The read operations are hence said to see the same view in both schedules.
- Condition 3 ensures that the final write operation on each data item is the same in both schedules, so the database state should be the same at the end of both schedules
- A schedule S is said to be view serializable if it is view equivalent to a serial schedule.

View Serializable

- **Constrained writes**: This condition states that any write operation $w_i(X)$ in T_i is preceded by a $r_i(X)$ in T_i depends only on the value of X read by $r_i(X)$.
- This assumes that computation of the new value of X is a function f(X) based on the old value of X read from the database.
- **Blind write**: The value written by $w(X)$ in T_i is independent of its old value, so it is not preceded by a read of X in the transaction T

T1	T2	T3
r1(x)		
w1(x)		
	$w2(X) -> Blindwrite$	
		$w3(X) -> Blindwrite$

Table 3: Example - Blind Write

View Serializable

T1	T2	T3
r1(x)		
w1(x)		
	$w2(X)$	
		$w3(X)$

Table 4: Example - Serial Schedule

T1	T2	T3
r1(x)		
	$w2(X)$	
w1(x)		
		$w3(X)$

Table 5: Example - Non - Serial Schedule

Check whether the schedule is view serializable and Conflict serializable?

References

Fundamentals of Database systems 7th Edition by Ramez Elmasri.